

1. Polynomial Spaces

Let $\mathcal{P}_n$ denote a real-valued vector space of polynomials of degree less than or equal to n , where n is a nonnegative integer. A generic polynomial in $\mathcal{P}_n$ can be expressed as follows:

$$p(t) = \sum_{k=0}^n p_k t^k = \underbrace{\begin{bmatrix} 1 & t & \cdots & t^n \end{bmatrix}}_{\mathbf{f}^\top(t)} \underbrace{\begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_n \end{bmatrix}}_{\mathbf{p}} = \mathbf{f}^\top(t) \mathbf{p},$$

where $t \in \mathbb{R}$, $\mathbf{f}(t) \in \mathbb{R}^{n+1}$ denotes the vector of monomials (you can think of it as a vector-valued function of t), $\mathbf{p} \in \mathbb{R}^{n+1}$ denotes the vector of the coefficients, and $^\top$ denotes the transpose.

Define $\mathbb{V} \subseteq \mathcal{P}_n$ as the subset of all polynomials in $\mathcal{P}_n$ that have $t = 0$ as a root. That is,

$$\mathbb{V} = \left\{ p(t) = \sum_{k=0}^n p_k t^k \mid p(0) = 0, p_k \in \mathbb{R}, k = 0, \dots, n \right\}.$$

- (a) Explain why $\mathbb{V}$ is a subspace of $\mathcal{P}_n$.

Answer:

The set $\mathbb{V}$ is nonempty because the zero polynomial $p(t) = 0$ for all $t \in \mathbb{R}$ is contained in $\mathbb{V}$.

Now, consider two polynomials in $\mathbb{V}$, given by $p(t) = p_0 + p_1 t + \cdots + p_n t^n$ and $q(t) = q_0 + q_1 t + \cdots + q_n t^n$. Let α and β be arbitrary real numbers, and define the polynomial $r(t) = \alpha p(t) + \beta q(t)$ for all $t \in \mathbb{R}$.

Expanding $r(t)$, we have:

$$r(t) = (\alpha p_0 + \beta q_0) + (\alpha p_1 + \beta q_1)t + \cdots + (\alpha p_n + \beta q_n)t^n.$$

Thus, $r(t)$ is also a polynomial of degree at most n , meaning $r(t)$ belongs to the space of polynomials of degree at most n , denoted $\mathcal{P}_n$.

Moreover, since both $p(t)$ and $q(t)$ satisfy $p(0) = 0$ and $q(0) = 0$, it follows that:

$$r(0) = \alpha p(0) + \beta q(0) = 0.$$

Hence, $r(t)$ satisfies the same condition and belongs to $\mathbb{V}$.

Since $\mathbb{V}$ is nonempty and closed under linear combinations, it follows that $\mathbb{V}$ is a subspace of $\mathcal{P}_n$.

Note that closure under linear combinations is equivalent to closure under scalar multiplication and closure under addition.

- (b) Show that each polynomial in $\mathcal{P}_n$ can be represented by a unique vector in $\mathbb{R}^{n+1}$. Similarly, show that each polynomial in $\mathbb{V}$ can be represented by a unique vector in $\mathbb{R}^n$.

Answer: Each polynomial in $\mathcal{P}_n$ uniquely corresponds to the vector $\begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_n \end{bmatrix} \in \mathbb{R}^{n+1}$, where $p_0, p_1 \dots p_n$

are the coefficients of the polynomial. We can see that this is true as if two polynomials are equal to each other, then their coefficients must match and thus will correspond to the same vector. Similarly, if we start with a vector in $\mathbb{R}^{n+1}$, then we get a unique polynomial in $\mathcal{P}_n$ by having the coefficients of the polynomial be equal to the entries of the vector.

For $\mathbb{V}$, we have that $p(t) = \sum_{k=0}^n p_k t^k \implies p(0) = p_0 = 0$. Besides the constant being 0, there are no other constraints, so the polynomial must be of the form $p(t) = p_1 t + \dots + p_n t^n$. Therefore each

polynomial in $\mathbb{V}$ uniquely corresponds to the vector $\begin{bmatrix} p_1 \\ \vdots \\ p_n \end{bmatrix} \in \mathbb{R}^n$ by using a similar argument used for $\mathcal{P}_n$.

2. Fun with Cauchy-Schwarz

The Cauchy-Schwarz inequality states that for two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| = |\mathbf{x}^\top \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$$

(a) Without using a calculator, show that

$$1(100) + 2(99) + 3(98) + \dots + 100(1) \leq 1^2 + 2^2 + 3^2 + \dots + 100^2$$

Answer: The key realization here is that we can represent these summations as vector inner products and norms. Let's define $\mathbf{u} = [1 \ 2 \ \dots \ 100]^\top$ and $\mathbf{v} = [100 \ 99 \ \dots \ 1]^\top$. Then the left-hand side of the inequality can be written as

$$\sum_{i=1}^{100} i(101-i) = \sum_{i=1}^{100} u_i v_i = \mathbf{u}^\top \mathbf{v} = |\mathbf{u}^\top \mathbf{v}|$$

We can also break up the right-hand side:

$$\sum_{i=1}^{100} i^2 = \left(\sqrt{\sum_{i=1}^{100} i^2} \right) \left(\sqrt{\sum_{i=1}^{100} i^2} \right) = \|\mathbf{u}\| \cdot \|\mathbf{v}\|$$

Using Cauchy-Schwarz, we know that $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \cdot \|\mathbf{v}\|$, which implies that $1(100) + 2(99) + 3(98) + \dots + 100(1) \leq 1^2 + 2^2 + 3^2 + \dots + 100^2$.

Further, note that we can actually make this a strict inequality because the two vectors are linearly independent. We can verify this by showing that there is no scalar α such that $\mathbf{u} = \alpha \mathbf{v}$. Therefore, since the two vectors are linearly independent, $\cos(\theta) \neq 1$ since $\theta \neq 0$, and $|\mathbf{u}^\top \mathbf{v}| < \|\mathbf{u}\| \cdot \|\mathbf{v}\|$.

(b) Show that for any $\mathbf{v} \in \mathbb{R}^n$ such that all entries $v_i \geq 0$

$$\|\mathbf{v}\|_1 \leq \sqrt{n} \|\mathbf{v}\|_2$$

HINT: Recall the following definition of the 1-norm, $\|\mathbf{v}\|_1 = \sum_{i=1}^n |v_i|$

Answer: We can massage the above equation into the form of Cauchy-Schwarz by reformulating the 1-norm as a vector product. Since we've been given that all $v_i \geq 0$, we can remove the absolute value from the definition of the 1-norm to say that

$$\|\mathbf{v}\|_1 = \sum_{i=1}^n v_i = \sum_{i=1}^n v_i(1) = \mathbf{v}^\top \mathbf{1} = |\mathbf{v}^\top \mathbf{1}|$$

where $\mathbf{1}$ is the vector in which each entry is one.

By Cauchy-Schwarz, we have

$$|\mathbf{v}^\top \mathbf{1}| \leq \|\mathbf{v}\|_2 \cdot \|\mathbf{1}\| \tag{1}$$

$$= \|\mathbf{v}\|_2 \cdot \left(\sqrt{\sum_{i=1}^n 1} \right) \tag{2}$$

$$= \sqrt{n} \|\mathbf{v}\|_2 \tag{3}$$

Therefore, $\|\mathbf{v}\|_1 \leq \sqrt{n} \|\mathbf{v}\|_2$.

BONUS: Can you prove that this inequality holds true for all $v_i \in \mathbb{R}$?